

Solaris: Designing for Unknowability in Interactive Systems

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Abstract. This paper presents Solaris, an interactive WebGL simulation of Lem’s sentient ocean, built around a single design question: can a simulation produce the experience of encountering something whose interior stays closed to the viewer? The work treats unknowability as a property of an encounter rather than of a system, and argues that authoring it requires ontological choices made in code—in physics, communication structure, and response grammar. Four design resources produce this refusal independently of system complexity—category defeat, hidden state, timescale mismatch, and contentless channels—examined through three subsystems: a viscoelastic physics model with memory and intent terms, a gravitational labor visualization, and a mood-governed response system. A fourth subsystem, the mimoid, fails: its generative principle operates on recognizable referents, and the encounter collapses the moment a viewer names one.

CCS Concepts. • Human-centered computing → Interaction design theory, concepts and paradigms; • Computing methodologies → Simulation types and techniques; • Applied computing → Media arts.

Keywords. interactive art, viscoelastic simulation, WebAssembly, generative systems, human-computer interaction, Stanisław Lem

Accepted Author Manuscript. The Version of Record is available from ACM at <https://doi.org/10.1145/3840416.3840469>. This author-prepared version differs in pagination and layout from the Version of Record and was not produced by ACM.

ACM Reference Format:

Magda Hamilton. 2026. Solaris: Designing for Unknowability in Interactive Systems. In *Expanded 2026: Conference on Animation and Interactive Art (Expanded 2026)*, September 09–11, 2026, Linz, Austria. ACM, New York, NY, USA, 5 pages. DOI: [10.1145/3840416.3840469](https://doi.org/10.1145/3840416.3840469)

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1 Introduction

The “alien” in interactive media is almost always exotic familiarity: strange forms covering recognizably human structures. Even procedural systems built around generative variety produce novelty within stable forms, because interactivity trains us to look for the loop that closes—the input that produces a readable output [1, 16]. A system the viewer cannot interpret tends to register not as alien but as broken.

Solaris, an interactive WebGL simulation of Lem’s sentient ocean [11], turns this problem into a design question: not how to make something look alien, but whether a simulation can produce the experience of encountering something whose interior stays permanently closed to the viewer.

The wager of the work is that this kind of alterity cannot be authored at the surface. Aesthetic strangeness wears off; behavioral mystery dissolves once the viewer constructs a theory of the system. Unknowability, the mechanism by which alterity is authored, has to be constructed at the level of logic—in physics, response grammar, and communication structure. Ontological choices made in code do interpretive work that visuals alone cannot.

The paper proceeds as follows. Section 2 situates the work against Lem’s source material, the philosophical question of radical Otherness, and adjacent practices in generative art. Section 3 describes what a viewer encounters. Section 4 presents three technical decisions as a single argument about authoring the unknowable. Section 5 turns to the mimoid system, where the design principles break down, and Section 6 draws out the resulting design vocabulary.

2 Context

Lem’s *Solaris* opens on a scientific facility orbiting a planet covered by a single colloidal organism—an ocean that is, by every measurable indicator, conscious. Generations of researchers have catalogued its behaviors and classified its formations, producing a vast body of literature—solaristics—that fills the station’s library. None of it has yielded an interpretation. This is not a failure of method or patience. Lem argues in *Summa Technologiae* that the situation is structural: communication with an intelligence whose cognitive architecture differs sufficiently from our own is not a problem awaiting a better protocol but an impossibility in principle [12].

The philosophical stakes are old. Wittgenstein grounds meaning in a shared form of life [21]; Nagel shows subjective experience to be structurally inaccessible across sufficiently different perceptual architectures [17]; Hayles treats our assumptions about what intelligence must look like as products of a specific embodied history [9]. Lem’s novel dramatizes what they converge on—an impossibility in principle that a simulation cannot reproduce but can stage.

Several works in computational art have refused legibility on individual layers. Jonathan McCabe’s multi-scale Turing patterns render reaction-diffusion computation as a material the viewer reads as living tissue yet cannot place [14]. Jenna Sutela’s *nimiia cétii* generates a more-than-human language whose script stays indecipherable [20]. Ian Cheng’s *Emissaries* trilogy runs autonomous agents that pursue internal agendas without observer input, though agents with goals of this kind invite working models quickly [4]. Each refuses

legibility on one dimension—material, communication, behavior—while remaining legible elsewhere. This work attempts refusal across every layer at once.

Interaction design has its own literature on withheld legibility. Gaver, Beaver and Benford propose ambiguity as a design resource [6]; Sengers and Gaver argue that multiple and even contradictory interpretations should be expected rather than corrected [19]; Hallnäs and Redström treat time as a material in designing for reflection [7]; Bell, Blythe and Sengers propose defamiliarization for making the familiar available to reconsideration [2]. This work departs from all four on one axis: ambiguity and defamiliarization operate on what the viewer already recognizes, and make the viewer a producer of interpretation. Solaris withholds the material for that production—no familiar object beneath the strangeness, no interpretation the viewer is invited to supply.

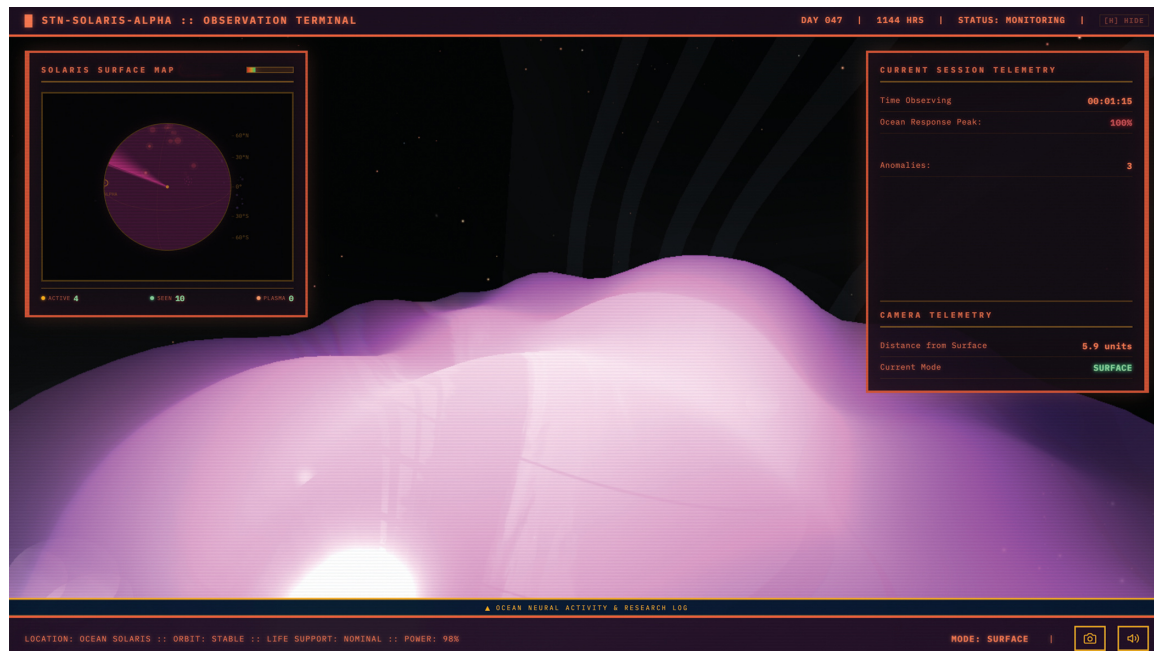
Generative art theory provides a further frame. Galanter locates generative systems on a continuum from order to disorder with effective complexity in between [5]; Boden and Edmonds distinguish generation that traverses a known space from generation that alters it [3]. Solaris attempts something orthogonal: not the production of novelty within or beyond a defined space, but the production of presence whose generative principle is not recoverable from observation.

These frames describe properties of systems, and the paper does not dispute that Solaris can be described as effectively complex. The distinction drawn here is a change of variable: unknowability, as used in this paper, is a property of an encounter—a relation between what a system does and what a viewer, within a given span of attention, can observe of it. What follows is therefore not a claim to greater complexity than this work's neighbors but an account of four design resources—category defeat (the thing fits no familiar perceptual category), hidden state (governing variables are never exposed), timescale mismatch (governing periods that never align within a session), and contentless channels (evidence of process without decodable product)—that produce refusal independently of it. Section 5 returns to these where the work fails to maintain them.

3 The Work

Solaris is a browser-based real-time simulation in which the viewer enters orbit above a planet whose entire surface is a single ocean, lit by two suns on different orbital periods. Camera movement is first-person; at close range it transitions to surface traversal. There is no objective and no end state. The ocean does not read as a planet, but as an organism. The surface pulses with the irregular rhythm of something breathing, color zones shifting and bleeding as the tidal field moves beneath them.

The cursor is the primary instrument of interaction. Hovering on the surface produces a localized response whose character varies on a timescale of fifteen to thirty seconds without notification. In informal observation sessions, a recurring pattern emerged: the viewer moves the cursor toward the surface, something responds, and the next gesture tests whether the response holds. It does not hold, or holds differently. These sessions are not treated as empirical validation but as qualitative indicators guiding iterative design; a controlled study remains open. Nor was the pattern universal: some viewers read the variability as malfunction, others as pseudorandom variation of a kind familiar from earlier



interactive work—outcomes the design targets but cannot be shown to prevent. Six formation types emerge and dissolve on independent schedules—symmetriads, asymmetriads, mimoids, vertebrids, extensors, and fungoids [8]—regardless of viewer input.

Sustained attention is a precondition the design cannot assume. The research frame is built to manufacture it: a mission brief, prior researchers' logs, instruments that report without explaining, and a log accumulating entries the viewer did not author. None of it makes the ocean more comprehensible. Curiosity requires a gap that appears closable [13]; the instrumentation stages closable gaps continuously while the ocean closes none. The frame is legible so that the ocean does not have to be.

4 Case Study: Authoring the Unknowable

The technical decisions in Solaris are interpretive choices made to answer a specific question: how do you author a system that is observable in continuous detail and yet refuses to resolve into a model the viewer can hold? This section presents three of those decisions and argues that they are not parallel features of the work but a single argument approached from three angles. Each addresses the same problem at a different layer of system.

Four terms sit close together. Unpredictability concerns the next state; alone, it reads as malfunction. Complexity concerns the system; a complex system is legible if its state is exposed. Ambiguity, in Gaver et al.'s sense [6], concerns meaning: the work withholds a settled interpretation so the viewer can supply one. Unknowability concerns the relation between system and observer: the system is lawful, its lawfulness is perceptible, and the law is not recoverable within the attention available. The viewer is neither co-author nor solver but a failed interpreter.

4.1 Ontological Physics

The ocean is modeled as a viscoelastic gel using a Kelvin-Voigt solver, not as water using the wave equation. Water in real-time graphics has a settled language; a water simulation, however stylized, sits inside a perceptual category the viewer already has. The constitutive relation $\sigma = E \cdot \varepsilon + \eta \cdot \partial\varepsilon/\partial t$ —stress equals elastic strain plus viscous resistance—describes a material that resists deformation, stores energy, and releases it through wobble rather than wave, implemented across a 128×128 grid in C++/WebAssembly at sixty hertz [8].

The surface that emerges does not behave like water. In its unmodified form it behaves like gel or soft tissue; a viewer with experience of either will reach for them and be substantially right. What removes that footing is the addition of two fields. Each cell carries a memory term and an intent term, and these—not the solver—do the work of separating the surface from the categories it invites.

The memory term is a slowly drifting rest position, the equilibrium toward which elastic restoring forces pull the cell. In a standard solver this is fixed and the surface relaxes back to flat. Here it is updated each frame as a leaky integrator of the cell's recent deformation history, with a time constant of roughly thirty seconds. A region deformed repeatedly develops a higher equilibrium and retains a faint topographical residue after the disturbance ceases. The surface does not return to the same shape, because the shape it returns to has itself changed.

The intent term operates through two coupled mechanisms. Ambient motion—sinusoidal functions at three timescales: a slow large-scale undulation, wandering pressure zones, and sparse seeded-random stress injections—drives velocity independently of any external force, with amplitudes scaled by the current mood state. The second activates on observation: when the cursor contacts the surface, the ocean injects a localized target height the cell then springs toward, with a force proportional to the gap and a strength that decays as the ocean changes its mind. Both mechanisms are deterministic given seed and history, but neither is derivable from anything the viewer can observe: the ambient field runs whether the room is empty or not, and the spring target is set by internal logic the viewer has no access to. The goal is not to produce complexity, but to construct behavior that cannot be reduced to observable causes. Cells push themselves; the surface moves on its own.

Memory means the ocean is shaped by what has happened to it; intent means it acts on its own behalf, though neither term models cognition. The surface invites familiar categories—jelly, protoplasm, tissue—but those categories stop predicting almost at once. Jelly returns to rest; this surface returns to a rest position that has itself shifted. Gel does not deform without being deformed; this one does. Refusal of comprehension begins here: this is category defeat, at the level of what the simulated thing is.

4.2 Communication as Visible Labor

The ocean does not communicate through symbols—no patterns, signals, or text the viewer is invited to decode. What it does is work, and the work is made visible.

The dual-sun system is the substrate. The simulated planet orbits a red giant and a blue sun at different angular velocities, a configuration whose stability is not guaranteed: the three-body problem admits no general analytic solution [18]. In Lem's novel, the ocean's

continued existence in a binary configuration is itself a mystery—something stabilizes the orbit, but the mechanism is opaque.

The simulation makes this conceit explicit in a limited sense. The planet's orbit is not actually unstable—it is on rails—and the ocean is not doing the stabilization work. What the simulation does is compute, each frame, the tidal potential $(3 \cos^2 \theta - 1)/2$ from both suns at every cell on the grid, sum them, and apply a small fraction (5%) as displacement. The remaining 95% is treated, by stipulation, as the work the ocean performs to hold itself in place. Cells where this stipulated counteraction is highest accumulate a stress value, exported as a texture and visualized in the fragment shader as warm subsurface glow.

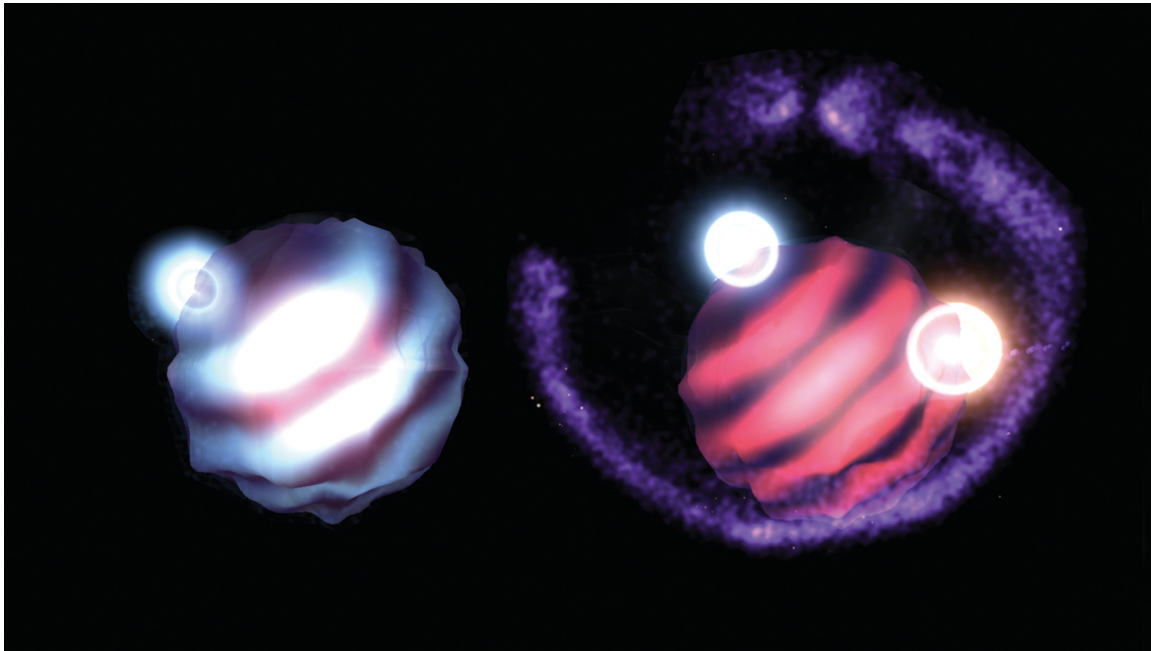


Figure 2. Tidal stress visualization. Aligned suns concentrate the glow (left); opposed suns diffuse it (right).

The stipulation produces a real perceptual artifact: regions of the surface luminescing in patterns that correspond, in real time, to gravitational forces the viewer cannot independently sense—sixty times per second. When the suns align and reinforce each other the glow concentrates and pulses; when they oppose, it diffuses. The patterns are neither arbitrary nor symbolic—they are the visible side of computation that is actually happening, bound to a fictional referent the viewer perceives only through its consequences. The ocean is not stabilizing anything; the arithmetic is ordinary. What differs is the encounter: the glow arrives without legend, on an entity already uncategorizable, and nothing converts intensity into meaning.

What this is designed to produce is the experience of watching effort. The ocean appears to be doing something difficult, and the difficulty is visible without being interpretable. The viewer can see that something is happening, where, and when it intensifies, but not what it is for. The channel is one-directional and contentless: it carries no product, though its correlation with sun position is observable. This matters because the alternative—symbolic

communication, however alien—invites decoding. A system that produces patterns invites the viewer to look for the cipher; a system that produces apparent labor offers nothing to decode.

4.3 Observation Without Legibility

The third layer is where the viewer enters. Every interactive system trains its viewer through legibility: input produces output, the viewer constructs a model, the model predicts further outputs, and the loop closes. This is the grammar of interactivity [1, 16]. The challenge is that the moment interaction becomes legible, the rest of the work's refusals read as theatrical—at least one channel has yielded.

The response system in *Solaris* is built to break this loop without breaking the work. Raycasting detects where the cursor meets the surface, feeding into the simulation as a disturbance whose character depends on a global mood state—curious, aloof, recoiling—that shifts every fifteen to thirty seconds along a schedule neither announced nor recoverable from observation. When curious, the surface lifts toward the cursor; when aloof, it flattens and dims; when recoiling, it withdraws. The transitions are not synchronized with anything the viewer does.

The result is response that is not random—randomness reads as broken—but is also not learnable. The viewer can see that hovering produces some effect; the viewer cannot determine which effect any given hover will produce. Standard interactive grammar requires that input map to output through a function the viewer can eventually approximate. This system maps input through a function whose parameters shift every fifteen to thirty seconds, governed by hidden logic the viewer cannot reconstruct within a single session. The function is real; it cannot be solved. It can be experienced, but not modeled.

What this performs, at the level of the viewer's experience, is the surface form of being studied by something whose interior is closed. The viewer reaches; the system responds; the response does not stabilize into a pattern; and the viewer cannot tell whether the variability is the system's mood, its attention, its indifference, or something else for which neither system nor viewer has a name.

The three decisions converge here. The ontology refuses comprehension because the substance is not familiar. The communication refuses because the channel carries no decodable content. The response refuses because the function is not learnable on the timescale of attention. None of these alone would be sufficient: strange physics with legible communication invites decoding, opaque communication with predictable response invites mastery, and unpredictable response with familiar physics reads as broken. Authored unknowability requires refusal at every constitutive layer simultaneously, because the viewer will use any available layer to construct the model the work is trying to refuse to deliver.

5 Structural Failure: The Mimoid Problem

The mimoid is one of the formation types listed in Section 3: a structure the ocean extrudes that imperfectly replicates objects from its surroundings. The novel's mimoids are dominated by what they get wrong—replicas partial, distorted, uncannily precise in some details

and grotesquely off in others. They are failing at mimicry, or doing something for which mimicry is the wrong word.

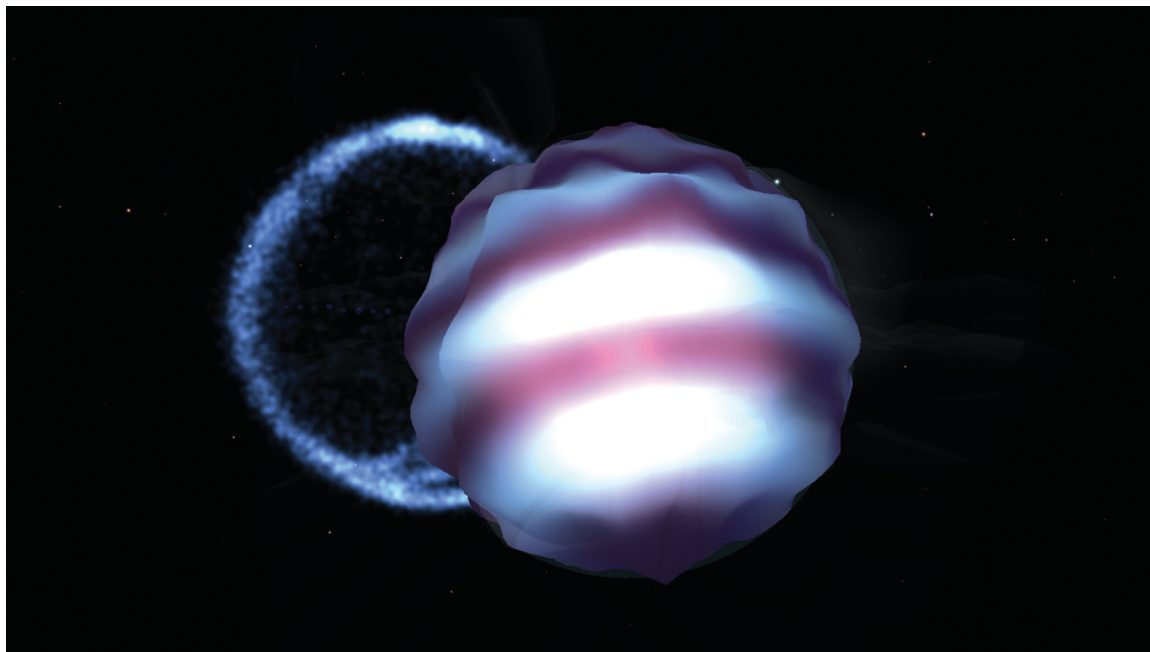


Figure 3. A mimoid approximating a sphere.

The implementation in *Solaris* attempts to capture this. The mimoid subsystem loads PLY mesh files and deforms their vertices in real time toward weighted mimicry targets: the observer’s silhouette, the suns, nearby formations, abstract geometric forms. Three layers of “wrongness”—temporal lag, memory contamination, FBM noise displacement—prevent perfect replication. The system produces formations that visibly try and fail to become other things.

What it does not do is sustain the refusal of comprehension that the rest of the work depends on. In the same informal observation sessions referenced in Section 3, viewers identified mimicry quickly: the sun being approximated, the body roughed in, the cube warped. Once the target was identified, the mystery collapsed and the system read as clever rather than uncanny—a structural contradiction in the design approach.

The diagnosis returns to Section 2. The other three layers refuse comprehension by defeating the categories they invite: the gel has no perceptual category, the labor has no decodable content, the response has no learnable function. The mimoid does the opposite. Its entire mechanism is comparison-with-known-things, and its comparison terminates. The viewer who sees the sun-shape says “ah, the sun,” and the loop closes: the work yields because the generative principle requires legibility to function at all. The novel’s own taxonomy—symmetriad, mimoid, vertebrid—supplies referents of a different kind: names that label without explaining, opening the question rather than closing it. Mori’s uncanny valley names one version of this tension: imperfect replication of recognizable referents reads as wrong before it reads as other, and once it reads as wrong, the interpretive apparatus locks onto the gap between target and execution [15]. What the rest of *Solaris*

tries to perform is not failed resemblance but unrecognizable presence—formations that do not invite comparison because there is nothing familiar enough to compare them to. The mimoid system, as currently built, operates in the wrong register.

Three directions present themselves, none yet tested: remove legible referents in favor of generation below the threshold of recognition; compress the lifecycle so the formation never resolves; or cut the mimoids. Each answers the same question: where does authored unknowability cross into theatrical unknowability, and what is the work willing to give up to stay on the right side of the line?

6 Contribution

The methodological claim this work advances is that alterity in interactive systems must be authored at the level of logic, not surface, and that the property being authored belongs to the encounter rather than to the system. Visual strangeness wears off; behavioral mystery collapses under interpretation; ontological refusal persists. This reframes interactive design from producing interpretable behavior to constructing systems that sustain non-interpretability through coherent logics that remain inaccessible from observation.

The four resources named in Section 2 are offered as a design vocabulary, addressed to designers building systems that resist mastery rather than reward it. They are not redundant, and do not decay at the same rate. Category defeat is spent; the others are not—the mood no more observable in the fortieth minute than in the first, the periods no closer to alignment, the glow no more decodable. The ontology buys the time in which their failure to resolve accumulates. The systems are deterministic; what matters is that a viewer cannot distinguish an unsolvable system from an unsolved one.

The mimoid problem sharpens the vocabulary. Recognition is not binary: viewers readily perceive behavioral qualities in systems with no physical correlate [10], and this perception is not a leak but a requirement for coherence. What collapses the encounter is recognition at the level of referent, where a formation resolves into a known thing and the question closes. Generative principles operating on referents are constitutively legible; principles operating on dynamics sustain coherence without conceding comprehension.

Since any layer that yields becomes an entry point for reduction, authored unknowability must be constructed across every layer that constitutes the encounter—not the scaffolding that sustains attention.

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